



Analog IC Design

LAB 05

Simple vs Wide Swing (Low Compliance)
Cascode Current Mirror

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1 Part 1: Exploring Sizing Trade-offs Using SA

Specs: sink current, $I_{in} = 10\mu A$, $I_{out} = 20\mu A$, % change in current for $\Delta V_{out} = 1V < 10\%$, mismatch $\sigma(I_{out})/I_{out} \leq 2\%$, compliance $\leq 150mV$, area minimized.

Q: Sinking current means which device type, NMOS or PMOS?

A: NMOS.

Q: What λ does the 10% spec translate to?

A: The output current changes as $\Delta I_{out} = \lambda I_D \Delta V_{OUT}$, so $\Delta I_{out}/I_{out} = \lambda \Delta V_{OUT}$. Setting this to 10% at $\Delta V_{OUT} = 1V$:

$$\lambda \leq \frac{0.10}{1V} = 0.1 \text{ V}^{-1} \quad (V_A \geq 10V)$$

and this has to hold at the biased point $V_{DS} = 0.5V$ since λ depends on V_{DS} .

1.1 The gm/ID trade-off table

Q: Fill in the trade-off table (higher gm/ID vs lower gm/ID).

A: everything except headroom wants a lower gm/ID (higher V^*):

Parameter	Higher gm/ID (lower V^*)	Lower gm/ID (higher V^*)
Dependence on V_{DS} (λ)	worse	better
Random mismatch	worse	better
Systematic mismatch	worse	better
Compliance voltage	better (small headroom)	worse (more headroom)

so raising V^* helps λ and mismatch and area, but the cost is headroom.

1.2 Pelgrom mismatch check

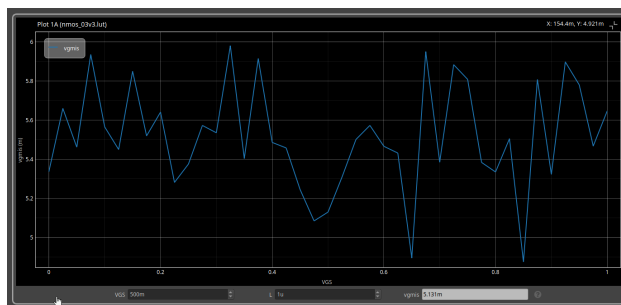


Figure 1: Vgmis vs VGS

Q: Find σ_{VT} from the model file (var.vth at $W \times L = 1\mu m^2$) and compare to ADT. Which is higher, why?

A:

```

.lib nfet_03v3.stat
subckt nfet_03v3_dss d g s b wibu l=0.25u nf=1 m=1 par=1 v_ssb=0.45u d_ssb=1.75u as=0 ps=0 pd=0 ncs=0 dtemp=0 sa=0 sb=0 sd=0
param par_vth=0.007148
param par_kn=0.007880
param par_lvth=36.7
param par_w=10.7
param par_leff=1.0
param par_weff=par_w*par_leff
param p_sqtarea=sqrt(par_leff*par_weff)
param var_kn=0.7071*par_kn*(1-0.001*par_sqtarea)
param mis_kn=var_kn*(1-0.001*par_sqtarea)
param var_vth=0.7071*par_vth*(1-0.001*par_sqtarea)
param mis_vth=var_vth*(1-0.001*par_sqtarea)
model nfet_03v3_dss nfet_03v3
v1 d d1 b nplus_u.m1 w=1u l=1 d_ssb=0 i1e-15 d_ssb
v2 s d1 b nplus_u.m1 w=1u l=1 d_ssb=0 i1e-15 d_ssb
m1 d1 g s d1 b nfet_03v3 ad=0 davar delvth= mis_vth*stat_mismatch l=1' own mulde=1 mis_kn*stat_mismatch nf=0 nf=0 ncs=0 pd=0 ps=0 sa=0 sb=0
sd=0 sd=0 w=1u
.ends

```

Figure 2: σ_{VT}

From the model file: $\text{par_vth} = 0.007148 \text{ (V}\cdot\mu\text{m)}$, so $A_{VT} = 7.148 \text{ mV}\cdot\mu\text{m}$. the mismatch of a single device is $\text{par_vth}/\sqrt{2}$. For a single transistor, $\sigma_{VT_model} = 0.007148 \times 0.7071 \approx 5.05\text{mV}$. From ADT, $\text{vgmis} = 5.131\text{mV}$. The ADT value is slightly higher because it's estimated from a finite Monte Carlo simulation, also maybe due to inaccuracies in calculating from random values but the main reason is that monte carlo takes into account all variations in all parameter not only Vth variations while the model files is only vth variations.

1.3 SA sweep at constant $\sigma(I_{out})/I_{out}$

Used the given formula $\frac{\sigma(I_{out})}{I_{out}} = \frac{\sqrt{1 + 1/m} \times idmis}{I_D} \times 100$ with $m = 2$ (output device has 2x area), and swept $V^* = 100\text{m}:200\text{m}$ in SA at $I_D = 10\mu\text{A}$, $V_{DS} = 0.5\text{V}$.

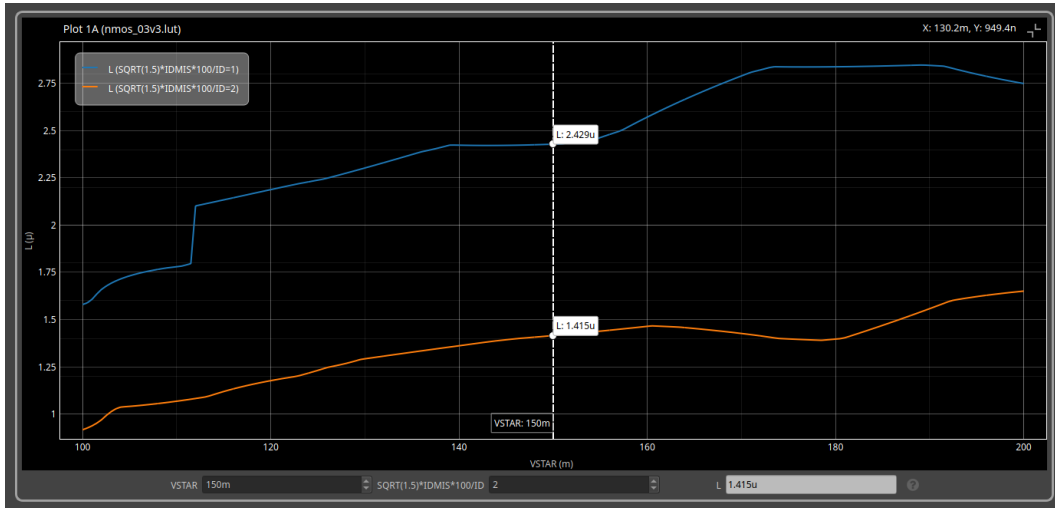


Figure 3: L vs V^* at $\sigma(I_{out})/I_{out} = 1\%$ (blue) and 2% (orange). Cursor at $V^* = 150\text{mV}$ gives $L = 1.415 \mu\text{m}$ for the 2% curve.

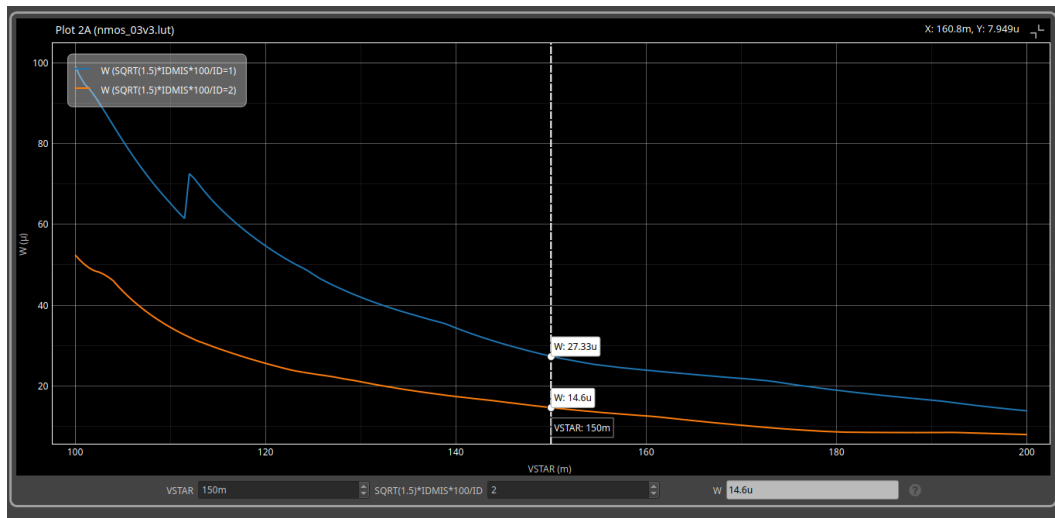


Figure 4: W vs V^* . Cursor at $V^* = 150\text{mV}$ gives $W = 14.6\text{ }\mu\text{m}$ for the 2% curve.

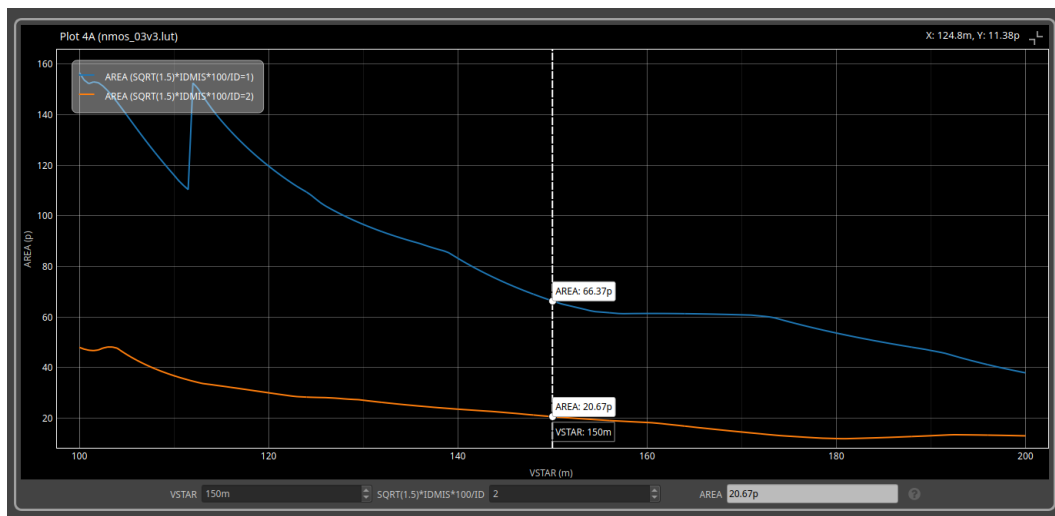


Figure 5: Area vs V^* . Cursor at $V^* = 150\text{mV}$ gives $\text{Area} = 20.67\text{p}$ (μm^2) for the 2% curve.

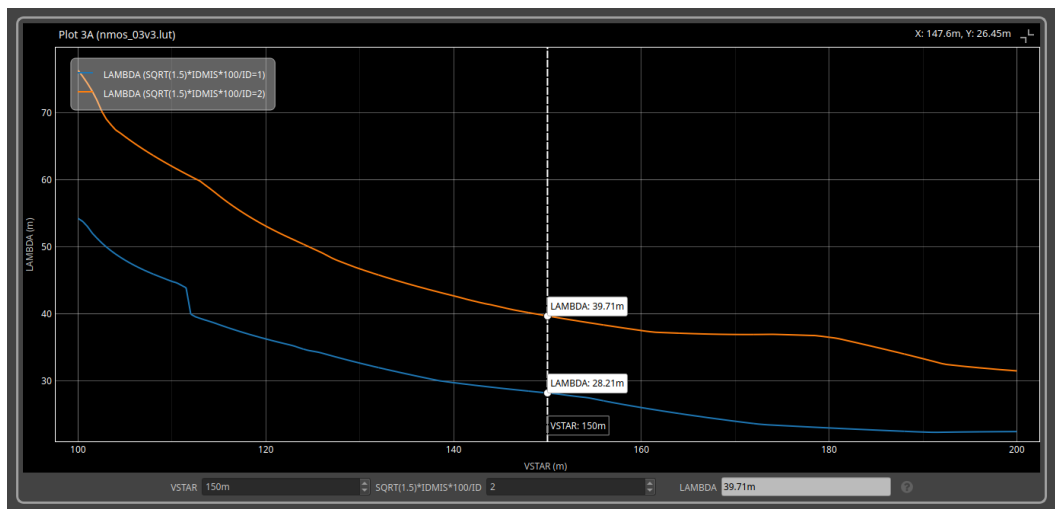


Figure 6: λ vs V^* . Cursor at $V^* = 150\text{mV}$ gives $\lambda = 39.71\text{m}$ for the 2% curve.

Q: Why does W get smaller and L get bigger as V^* goes up, and why does the area also shrink?

A: **Aspect ratio:** from $I_D = \frac{1}{2}\mu C_{ox} \frac{W}{L} V_{OV}^2$, for fixed I_D , a bigger $V^* = V_{OV}$ means you need a smaller W/L to hit the same current, so $W/L \propto 1/V^{*2}$. Since $W = \sqrt{(W/L) \cdot \text{Area}}$ and $L = \sqrt{\text{Area}/(W/L)}$, and the aspect ratio W/L shrinks much faster than the area does, W drops faster. while L actually grows.

Q: Why does λ decrease as L increases?

A: $V_A = \text{const} \times L$, so $\lambda = 1/V_A \propto 1/L$.

Q: Does the highest V^* look best for mismatch, area, and λ all at once? If so what limits it?

A: Yes, all three plots of mismatch spec already fixed at 2%, area, λ keep improving as V^* goes up, there's no trade-off between them. The only thing stopping us from just cranking V^* way up is the compliance voltage spec ($\leq 150\text{mV}$), since compliance $\approx V_{DSsat} \approx V^*$.

Q: Report the plots with a cursor at the required V^* . Does this point satisfy mismatch and λ ?

A: At $V^* = 150\text{mV}$: $L = 1.415\mu\text{m}$, $W = 14.6\mu\text{m}$, $\text{Area} = 20.67\text{p}$, $\lambda = 39.71\text{m/V}$. Mismatch is satisfied. λ spec was $\leq 100\text{m/V}$ and we got 39.71m/V .

Q: the final device sizing, λ , and $\sigma(I_{out})/I_{out}$ at the selected design point.

A: Final Part 1 design: $L = 1.42\mu\text{m}$, $W = 14.6\mu\text{m}$ (matches what SA reports, see below), $\lambda = 39.71\text{m/V}$, $\sigma(I_{out})/I_{out} = 2\%$ (by construction).

RESULTS:		
	Name	TT-27.0
1	ID	10u
2	IG	N/A
3	L	1.42u
4	W	14.6u

Figure 7: SA sizing result at $I_D = 10\mu\text{A}$, $V^* = 150\text{m}$, $V_{DS} = 0.5$, $\text{stack} = 1$: $L = 1.42\mu\text{m}$, $W = 14.6\mu\text{m}$.

2 Part 2: Current Mirror Simulation

2.1 Design and OP Analysis

$R_B \approx V_{DS2}/I_B$ with $V_{DS2} \approx V^* + 50\text{mV} = 150\text{m} + 50\text{m} = 0.200\text{V}$ and $I_B = 10\mu\text{A}$:

$$R_B \approx \frac{0.200\text{ V}}{10\mu\text{A}} = 20\text{ k}\Omega$$

just a starting point to pick a sweep range.

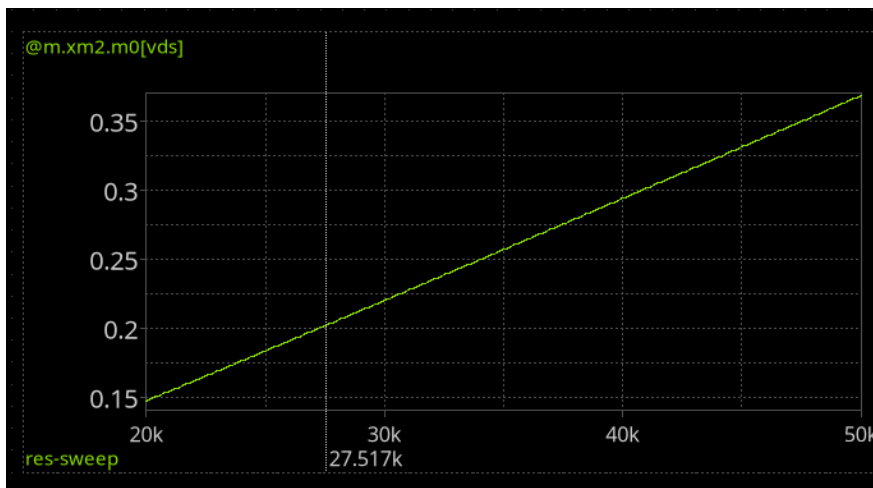


Figure 8: DC sweep of R_B from 20k to 50k, plotting V_{DS3} . Target crossing at $V_{DS3} = 0.2\text{V}$ happens around $R_B \approx 27.5\text{k}\Omega$.

Is it bigger or smaller than the rough value, why? It's **bigger** than the rough 20k estimate. The rough formula dropped the $V_{GS4} - V_{GS2}$ term: $R_B = (V_{GS4} + V_{DS2} - V_{GS2})/I_B \approx V_{DS2}/I_B$. but $V_{GS4} \neq V_{GS2}$ because M4 sits at a different bias point and suffers from body effect, which makes its V_{TH} and V_{GS} bigger.

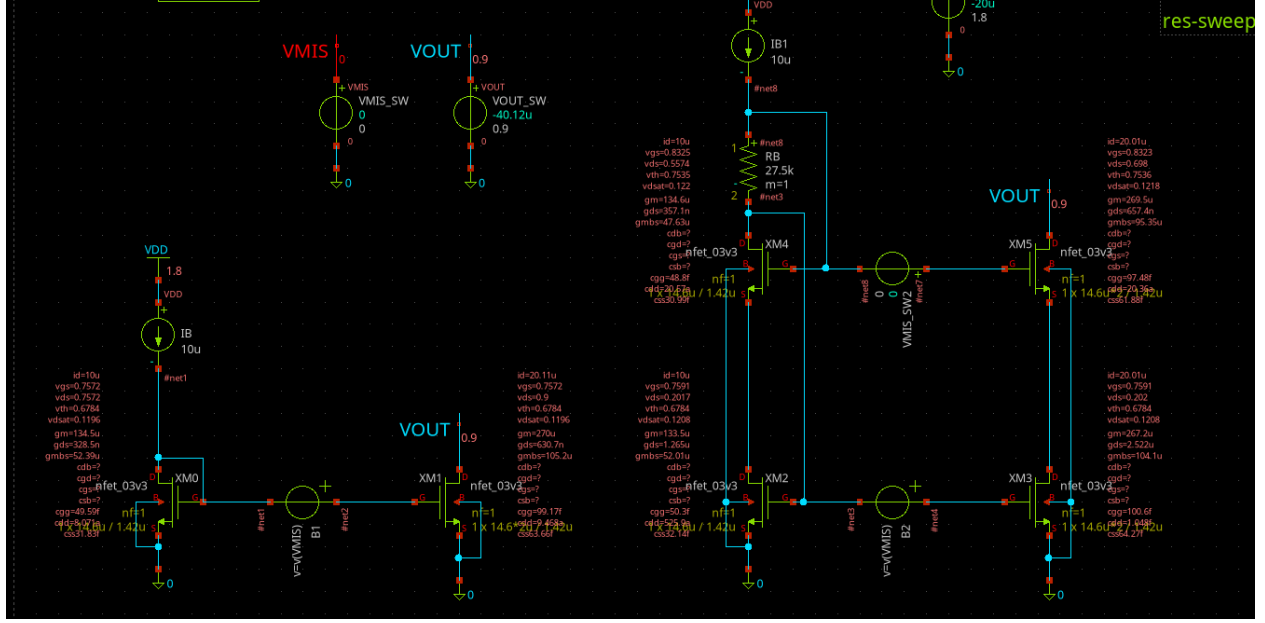


Figure 9: Full schematic with OP annotations for all six transistors (M0/M1 = simple mirror, M2/M3/M4/M5 = wide-swing/cascode mirror), $R_B = 27.5k\Omega$.

Dev	ID	VGS	VDS	VTH	VDSAT	GM	GDS
M0 (simple, ref)	$10\mu A$	0.7572	0.7572	0.6784	0.1196	134.6μ	$328.5n$
M1 (simple, out)	$20.11\mu A$	0.7572	0.9	0.6784	0.1196	270μ	$630.7n$
M2 (WS, ref-bottom)	$10\mu A$	0.7591	0.2017	0.6784	0.1208	133.5μ	1.265μ
M3 (WS, out-bottom)	$20.01\mu A$	0.7591	0.202	0.6784	0.1208	267.2μ	2.522μ
M4 (WS, ref-cascode)	$10\mu A$	0.8325	0.5574	0.7535	0.122	134.6μ	$357.1n$
M5 (WS, out-cascode)	$20.02\mu A$	0.8323	0.698	0.7536	0.1218	269.5μ	$657.4n$

Table 1: OP parameters read off ngspice (units: V, A, S). GMB: M0=52.39u, M1=105.2u, M2=52.01u, M3=104.1u, M4=47.63u, M5=95.35u.

Do all transistors operate in saturation? Yes, V_{DS} is bigger than V_{DSAT} , and the tightest ones (M2: $0.2017 > 0.1208$, M3: $0.202 > 0.1208$) which makes sense since we specifically designed R_B to give those two the 50mV margin above V_{DSAT} .

2.2 DC Sweep (I_{out} vs V_{OUT})

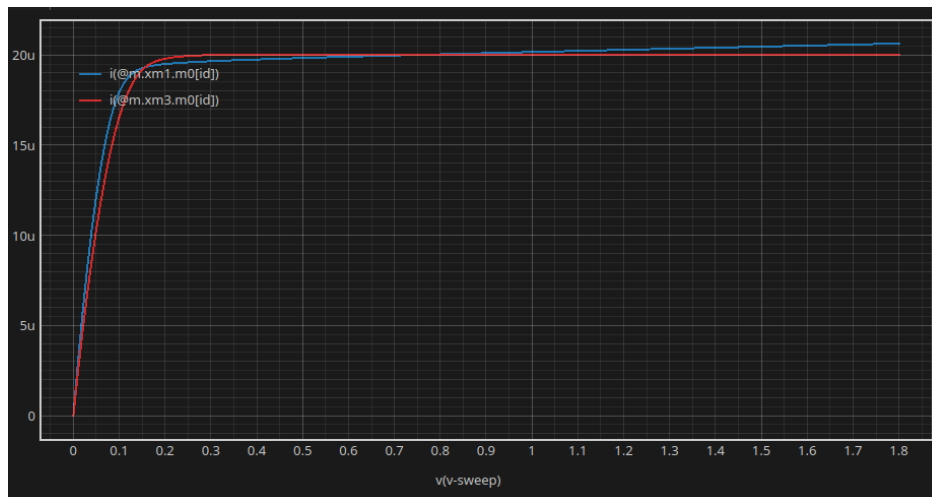


Figure 10: I_{out} vs V_{OUT} , simple CM (blue, xm1) vs wide-swing CM (red, xm3), V_{OUT} swept 0 to 1.8V.

Comment. Estimate compliance voltage. Why does the simple CM hit exactly $IB*2$ at one specific V_{OUT} ? The wide-swing (red) curve goes up to $20\mu A$ and stays flat the rest of the way, while the simple mirror (blue) keeps climbing steadily all the way to 1.8V – that’s the channel-length-modulation sensitivity we designed against showing up directly. **Compliance estimate:** simple CM: $V_{comp1} = 150mV$; wide-swing CM: $V_{comp2} = 300mV$. I_{out} of the simple mirror equals exactly $2I_B$ when $V_{DS1} = V_{DS0}$, when $V_{OUT} = V_{GS0} = 0.7572V$ at that point M0 and M1 have identical V_{GS} and V_{DS}

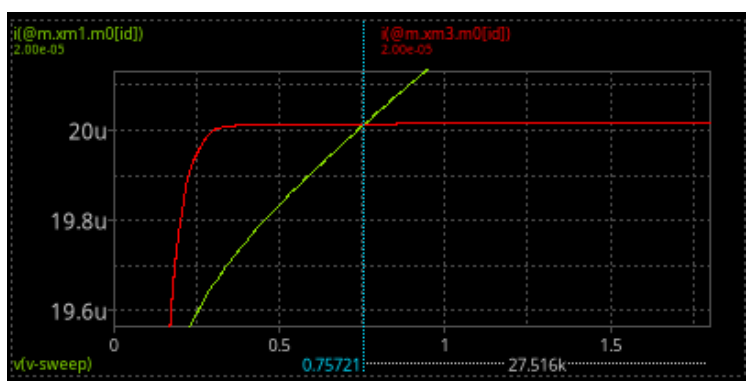


Figure 11: $V_{OUT} = 0.7572V \Rightarrow I_{out,simple} = 2I_B$ exactly, matching V_{GS0} from the OP table.

```

No. of Data Rows : 181
Simple Current Mirror Simulation Results
i1          = 1.983370e-05
i2          = 2.046595e-05
perc_change = 3.137744e+00
Wide Swing Current Mirror Simulation Results
i1          = 2.001215e-05
i2          = 2.001680e-05
page: no such command available in ngspice
perc_change = 2.323318e-02
binary raw file "cmirror_dc.raw"
ngspice 1 -> █

```

Figure 12: ngspice meas output: simple CM %change = 3.14%, wide-swing CM %change = 0.0232%.

Hand estimate using $\lambda = 39.71\text{m/V}$ from Part 1:

$$\frac{\Delta I_{out}}{I_{out}} \approx \lambda \Delta V_{OUT} = 0.03971 \times 1 = 3.97\%$$

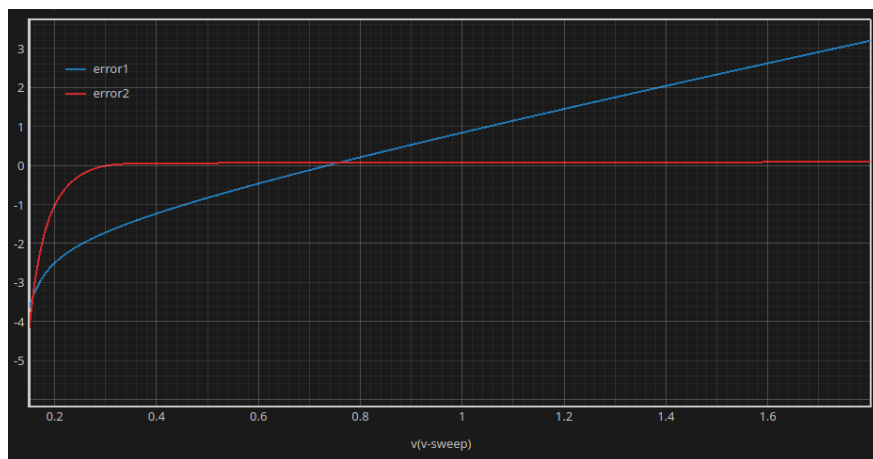


Figure 13: % error in I_{out} vs V_{OUT} (ideal = $2I_B$): error1 = simple CM (blue), error2 = wide-swing CM (red).

The simple mirror's error (blue) swings from about -4% at low V_{OUT} up to $+3.3\%$ at $V_{OUT}=1.8\text{V}$, crossing zero right at $V_{OUT} = 0.7572\text{V}$, since that's the exact-match point. The wide-swing mirror's error (red) snaps to basically 0% almost right after its compliance voltage.

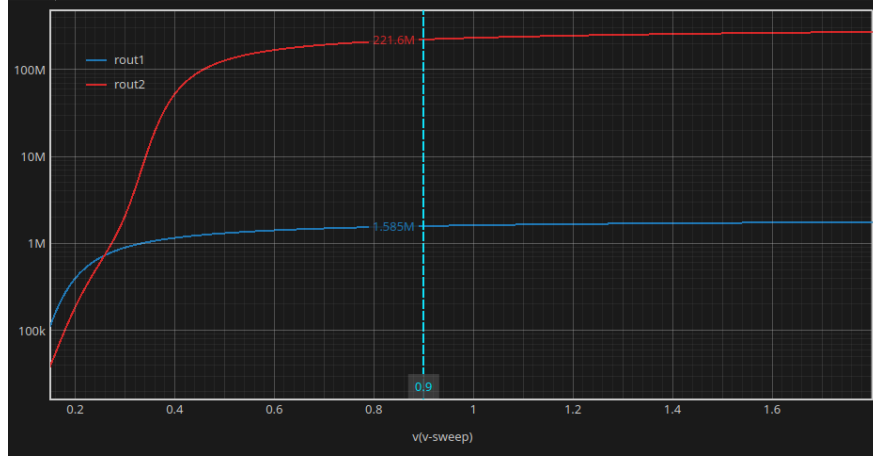


Figure 14: R_{out} vs V_{OUT} (log scale), simple CM (rout1, blue) vs wide-swing CM (rout2, red), cursor at $V_{OUT}=V_{DD}/2=0.9V$.

Does R_{out} change with V_{OUT} , why? At $V_{OUT} = 0.9V$: $R_{out,simple} = 1.585M\Omega$, $R_{out,WS} = 221.6M\Omega$ – the wide-swing mirror’s output resistance is much bigger, Both curves rise steeply at low V_{OUT} which’s the triode-to-saturation transition. R_{out} does still drift slightly upward with V_{OUT} even in saturation because λ itself slowly decreases as V_{DS} increases so r_o increases. **Analytically**

calculate R_{out} of both circuits at $V_{OUT}=V_{DD}/2$, compare to simulation. Simple mirror: r_o ,

$$R_{out,simple} = \frac{1}{g_{ds1}} = \frac{1}{630.7n} = 1.585M\Omega$$

which matches the simulated $1.585M\Omega$ exactly.

Wide-swing (cascode) mirror: standard cascode output resistance formula,

$$R_{out,WS} = r_{o5} + r_{o3} + (g_{m5} + g_{mb5}) r_{o5} r_{o3}$$

with $r_{o3} = 1/g_{ds3} = 1/2.522\mu = 396.5k\Omega$, $r_{o5} = 1/g_{ds5} = 1/657.4n = 1.521M\Omega$, $g_{m5} + g_{mb5} = 269.5\mu + 95.35\mu = 364.85\mu A/V$:

$$R_{out,WS} = 1.521M + 0.3965M + (364.85\mu)(1.521M)(396.5k) \approx 221.8M\Omega$$

vs simulated $221.6M\Omega$ – basically a perfect match (0.1% off).

Circuit	Hand calc	Simulation
Simple CM	1.585 M Ω	1.585 M Ω
Wide-swing CM	221.8 M Ω	221.6 M Ω

Table 2: R_{out} hand calc vs simulation at $V_{OUT}=0.9V$.

2.3 Mismatch

```

No. of Data Rows : 18
Simple Current Mirror Simulation Results
i1      = 2.010552e-05 at= 0.000000e+00
i2      = 2.056807e-05 at= 1.700000e-03
perc_change = 2.274449e+00
Wide Swing Current Mirror Simulation Results
i1      = 2.001434e-05 at= 0.000000e+00
i2      = 2.046879e-05 at= 1.700000e-03
perc_change = 2.245133e+00
binary raw file "cmirror_dc.raw"
Doing analysis at TEMP = 27.000000 and TNOM = 27.000000

Using SPARSE 1.3 as Direct Linear Solver

No. of Data Rows : 18
"Wide Swing Current Mirror Simulation Results (Cascode
mismatch)": no such command available in ngspice
i1      = 2.001434e-05 at= 0.000000e+00
i2      = 2.001746e-05 at= 1.700000e-03
perc_change = 1.558761e-02
binary raw file "cmirror_dc.raw"
ngspice 1 -> █

```

Figure 15: ngspice output for the manual mismatch sweeps: VMIS_SW (mirror devices, both CMs) and VMIS_SW2 (cascode devices, WS only).

Simple CM: 2.274%. Wide-swing CM: 2.245%

Q: Analytically calculate that % change and compare.

A: Treating the mismatch voltage as a small-signal input to the mirror device (it behaves like a simple CS amplifier), $\Delta I_{out} = g_m \Delta V_{GS}$, so $\Delta I_{out}/I_{out} = (g_m/I_D) \Delta V_{GS}$. The sweep range used was $\Delta V_{GS} = 1.7\text{mV}$. For M1 (simple CM output): $g_m/I_D = 270\mu/20.11\mu = 13.43\text{V}^{-1}$

$$\frac{\Delta I_{out}}{I_{out}} = 13.43 \times 1.7\text{m} = 2.28\%$$

close to the simulated 2.274%. For M3 (WS CM output): $g_m/I_D = 267.2\mu/20.01\mu = 13.35\text{V}^{-1}$

$$\frac{\Delta I_{out}}{I_{out}} = 13.35 \times 1.7\text{m} = 2.27\%$$

also close to the simulated 2.245%.

Q: for VMIS_SW=0 and sweep VMIS_SW2 (cascode devices) instead. Find % change.

A: From the ngspice output: 0.01559% .

Q: Analytically calculate that % change and compare.

A: it as a *degenerated* CS amplifier: M5's gate sees the mismatch voltage, but its source is degenerated by r_{o3}

$$G_m = \frac{g_{m5}}{1 + (g_{m5} + g_{mb5}) r_{o3}}$$

$g_{m5} = 269.5\mu$, $g_{mb5} = 95.35\mu$, $r_{o3} = 396.5\text{k}\Omega$:

$$(g_{m5} + g_{mb5}) r_{o3} = 364.85\mu \times 396.5\text{k} = 144.7, \quad G_m = \frac{269.5\mu}{145.7} = 1.850\mu\text{A/V}$$

$$\frac{\Delta I_{out}}{I_{out}} = \frac{G_m \times 1.7\text{m}}{20.02\mu} = \frac{3.146\text{n}}{20.02\mu} = 0.0157\%$$

almost exactly the simulated 0.01559%

Q: Which mismatch contribution is more pronounced, why?

A: The mirror-device mismatch ($\approx 2.25\%$) completely bigger than cascode-device mismatch ($\approx 0.016\%$). That's because the cascode device's gate error gets attenuated by the source-degeneration factor $(1 + (g_m + g_{mb})r_o) \approx 146$, while the mirror device's gate error goes straight into $\Delta I = g_m \Delta V$ with none of that suppression.

Q: For a given total area, is it better to size the mirror and cascode devices equally, or give more area to the mirror devices?

A: Give more area to the mirror devices (M2/M3). Since the mirror mismatch is the dominant error source by a huge margin, and Pelgrom says $\sigma_{VT} \propto 1/\sqrt{WL}$, spending extra area on the cascode devices (M4/M5) gives almost nothing because it's degenerated by $RS=ro3$. So for a fixed area, oversize the bottom (mirror) pair and leave the top (cascode) pair minimum-sized.

2.4 Monte Carlo Simulation

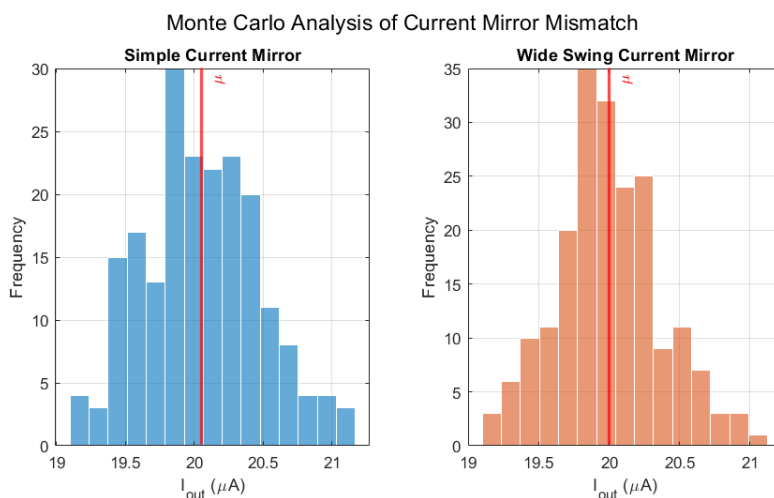


Figure 16: I_{out} histograms over 200 MC runs, simple CM (left) vs wide-swing CM (right), red line = mean.

```

--- Simple Current Mirror ---
Total Samples: 200
Average Iout: 20.0562 uA
Standard Deviation: 0.4210 uA
Percentage Error: 2.10%

--- Wide Swing Current Mirror ---
Total Samples: 200
Average Iout: 20.0002 uA
Standard Deviation: 0.3726 uA
Percentage Error: 1.86%

```

Figure 17: MC statistics: mean, standard deviation and % error (σ/I_{out}) for both mirrors.

Q: Calculate $\sigma(I_{out})/I_{out}$ from the MC run and compare to the expected analytical result.

A:

Circuit	Mean I_{out}	σ	σ/I_{out}
Simple CM	20.0562 μA	0.4210 μA	2.10%
Wide-swing CM	20.0002 μA	0.3726 μA	1.86%

Table 3: Monte Carlo results, 200 samples.

Both numbers land close to the 2% target we designed for in Part 1 ($\sigma(I_{out})/I_{out} \leq 2\%$ at the chosen $V^* = 150\text{mV}$ design point) – the simple CM’s 2.10% is basically right on spec, and the wide-swing CM comes out slightly better at 1.86%, which fits with what we found in the mismatch section: since cascode-device mismatch barely matters, the wide-swing mirror’s total mismatch is dominated by the same mirror-pair mismatch as the simple CM, so we’d expect them to be close.